

ВНЕВПИСАННАЯ ОКРУЖНОСТЬ

$$\alpha) AL = \frac{1}{2}P$$

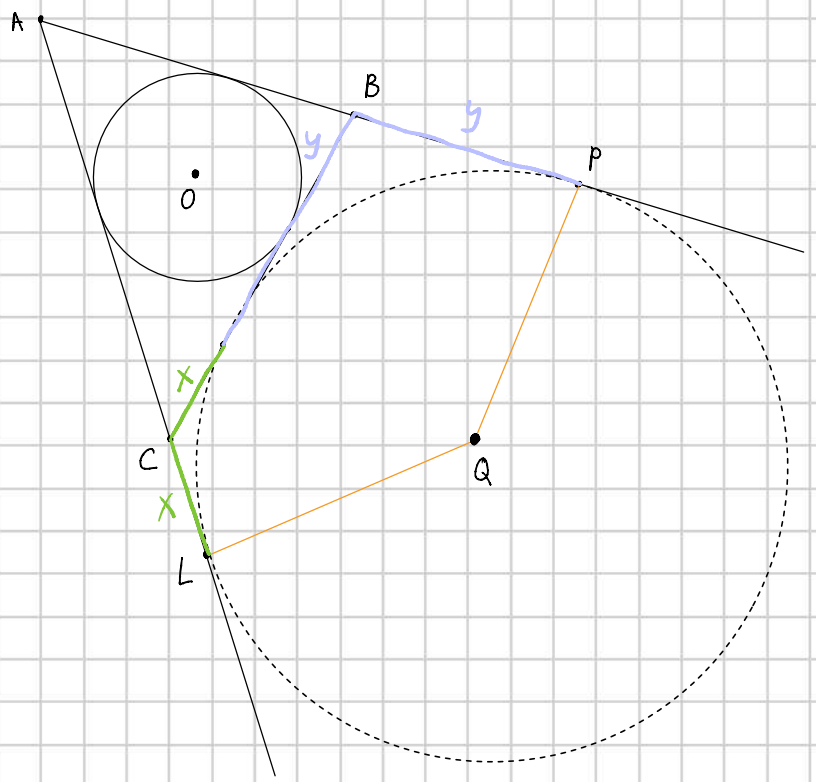
$$AL = AC + CL$$

$$AP = AB + BP$$

$$AP + AL = AB + AC + CL + BP = AB + AC + BC$$

$$2AL = P$$

$$AL = \frac{P}{2}$$

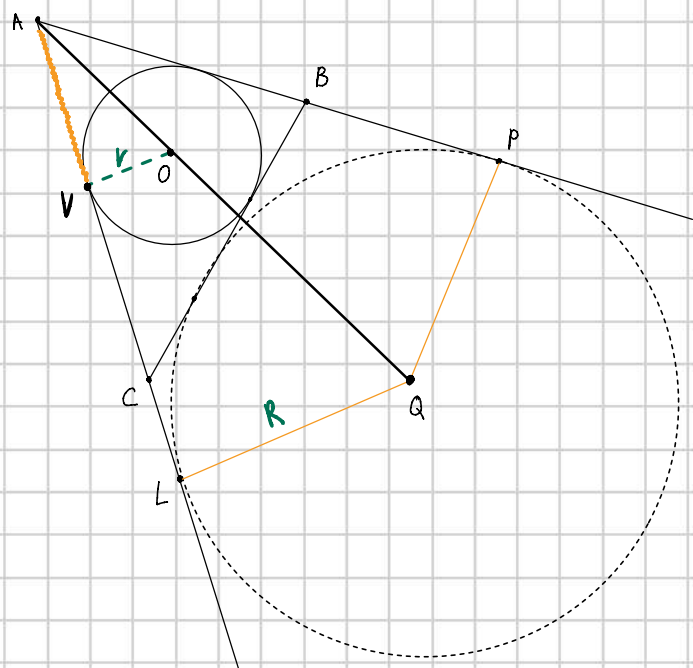


$$\beta) S_{\Delta} = (p - BC) \cdot R$$

$$S_{\Delta} = p \cdot r$$

$$\Delta AVO \sim \Delta ALQ :$$

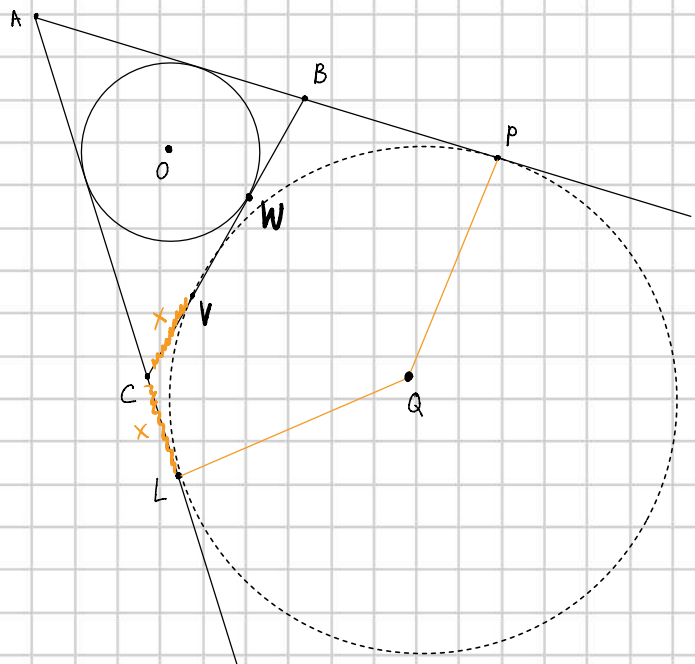
$$\frac{p - BC}{p} = \frac{r}{R} ; p \cdot r = (p - BC) \cdot R$$



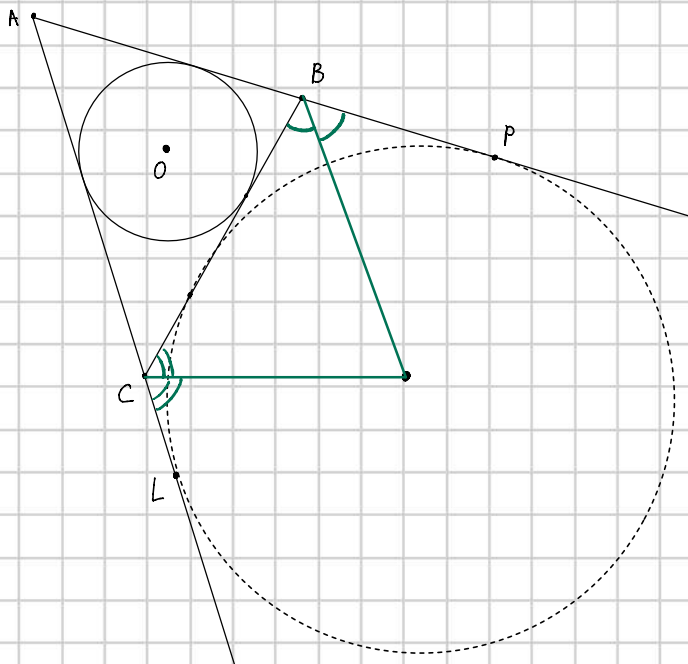
$$\gamma) CV = WB$$

$$CV = AL - AC = p - AC$$

$$WB = p - AC$$



$$\delta) \text{ БИСЕКТРИСЫ ПО ОПР.}$$

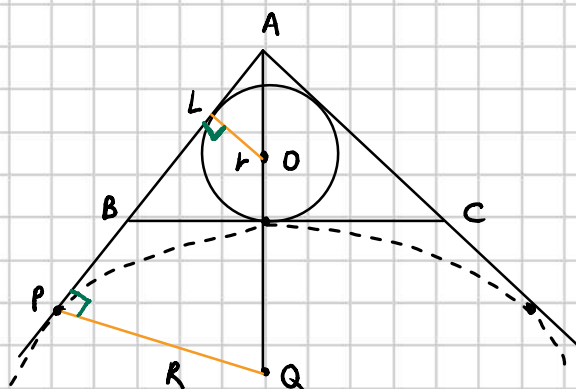


$$\epsilon) \text{ В РАВН. } \Delta : CB = 2\sqrt{Rr}$$

$$PL = 2\sqrt{Rr}$$

$$AL = AP - PL = p - 2\sqrt{Rr}$$

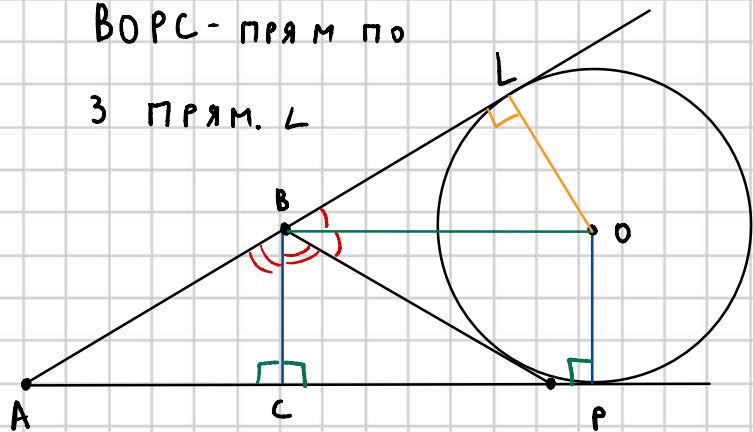
$$AL = p - BC$$



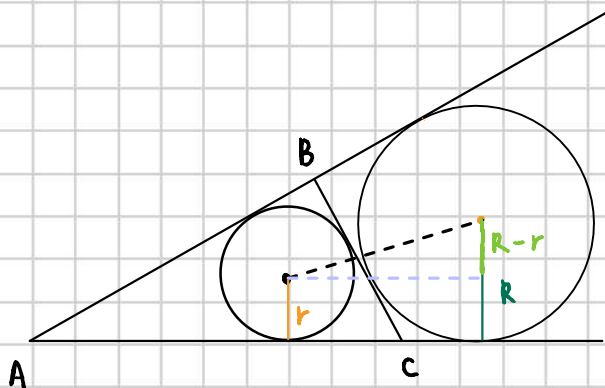
$$5) R = h \text{ (равн. } \Delta \text{)}$$

ВОРС - прям по

3 прям. L



$$7) R - r$$



$$8) k = \frac{r}{R_1}$$

$$S_{\Delta AOD} = \frac{1}{2} \cdot R_1 \cdot p$$

$$\frac{S_{\Delta AOS}}{S_{\Delta AOD}} = k^2; \quad \frac{2S_{\Delta AOS}}{R_1 \cdot p} = k^2;$$

$$2S_{\Delta AOS} = \frac{r^2 \cdot p}{R_1} = \frac{S_{\Delta ABC}}{R_1} \cdot r$$

$$S_{\Delta ABC} = \left(\frac{S_{\Delta ABC}}{R_1} + \frac{S_{\Delta ABC}}{R_2} + \frac{S_{\Delta ABC}}{R_3} \right) \cdot r$$

$$\frac{1}{r} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

